Machine Learning Assignment :- 13

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Q. Implement K-Means clustering on Iris.csv dataset. Determine the number of clusters using the elbow method.

DATASET LINK :- https://www.kaggle.com/datasets/uciml/iris

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
```

Dataset

```
iris = load_iris()
dataset = pd.DataFrame(data=iris.data, columns=iris.feature_names)
dataset.head()
dataset.info()
dataset.describe()
dataset.isnull().sum()
```

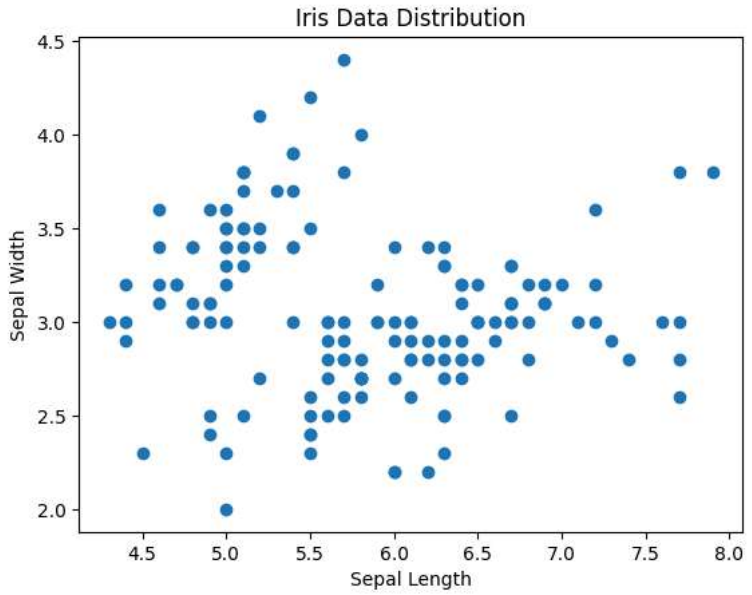
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 4 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   sepal length (cm)  150 non-null    float64
 1   sepal width (cm)   150 non-null    float64
 2   petal length (cm)  150 non-null    float64
 3   petal width (cm)   150 non-null    float64
dtypes: float64(4)
memory usage: 4.8 KB
```

	0
sepal length (cm)	0
sepal width (cm)	0
petal length (cm)	0
petal width (cm)	0

dtype: int64

Data Scatter Plot

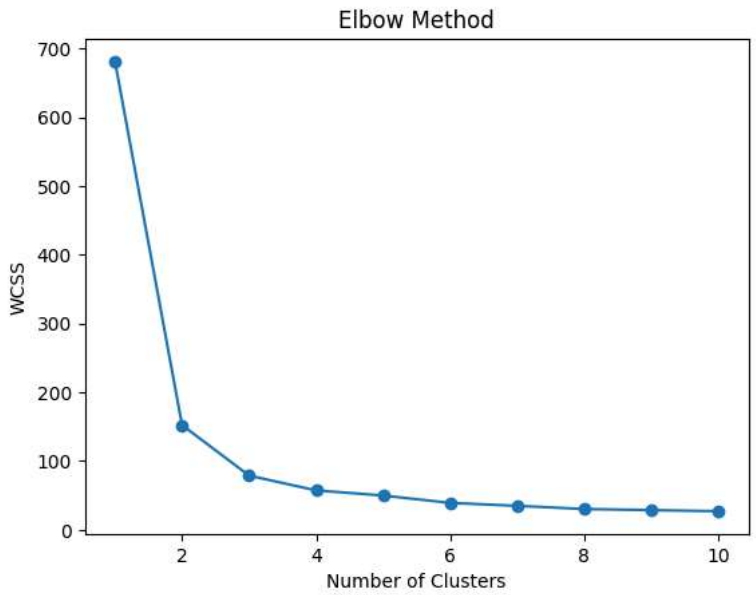
```
plt.scatter(dataset.iloc[:, 0], dataset.iloc[:, 1])
plt.xlabel("Sepal Length")
plt.ylabel("Sepal Width")
plt.title("Iris Data Distribution")
plt.show()
```



Elbow Method

```
X = dataset.values
wcss = []
for i in range(1,11):
  kmeans = KMeans(n_clusters=i, init='k-means++', random_state=17)
  kmeans.fit(X)
  wcss.append(kmeans.inertia_)

plt.plot(range(1,11), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()
```



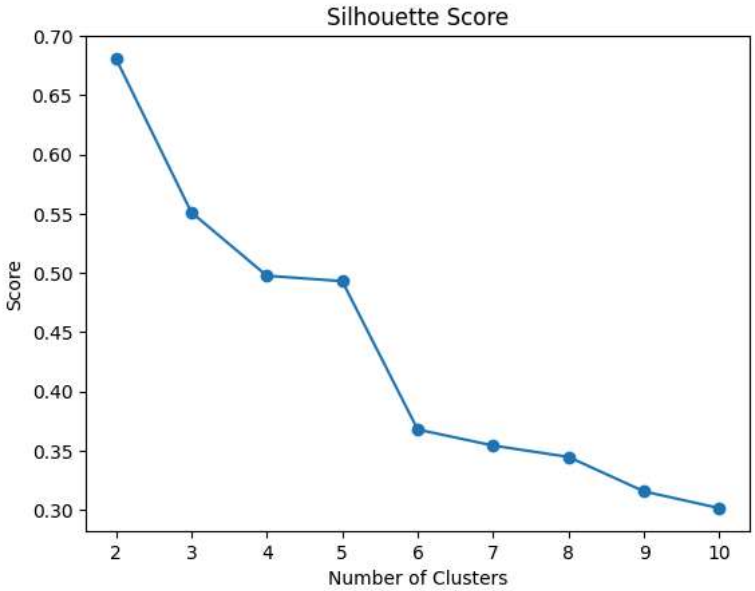
Silhouette Score

```
sil_scores = []
for i in range(2,11):
  kmeans = KMeans(n_clusters=i, init='k-means++', random_state=42)
  labels = kmeans.fit_predict(X)
  score = silhouette_score(X, labels)
  sil_scores.append(score)
```

```
    print(f"K = {i}, Silhouette Score = {score:.4f}")

plt.plot(range(2,11), sil_scores, marker='o')
plt.title("Silhouette Score")
plt.xlabel("Number of Clusters")
plt.ylabel("Score")
plt.show()
```

```
K = 2, Silhouette Score = 0.6810
K = 3, Silhouette Score = 0.5512
K = 4, Silhouette Score = 0.4976
K = 5, Silhouette Score = 0.4931
K = 6, Silhouette Score = 0.3678
K = 7, Silhouette Score = 0.3543
K = 8, Silhouette Score = 0.3447
K = 9, Silhouette Score = 0.3156
K = 10, Silhouette Score = 0.3014
```

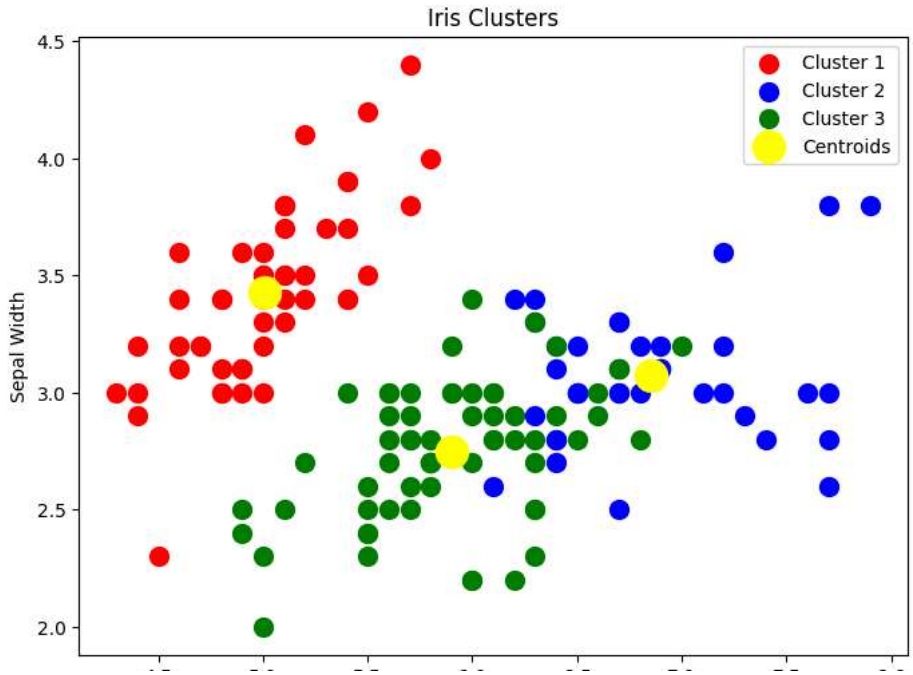


Clustering

```
kmeans = KMeans(n_clusters=3, init='k-means++', random_state=12)
y_kmeans = kmeans.fit_predict(X)

plt.figure(figsize=(8,6))
plt.scatter(X[y_kmeans == 0, 0], X[y_kmeans == 0, 1], s=100, c='red', label='Cluster 1')
plt.scatter(X[y_kmeans == 1, 0], X[y_kmeans == 1, 1], s=100, c='blue', label='Cluster 2')
plt.scatter(X[y_kmeans == 2, 0], X[y_kmeans == 2, 1], s=100, c='green', label='Cluster 3')

plt.scatter(kmeans.cluster_centers_[:,0], kmeans.cluster_centers_[:,1],
s=300, c='yellow', label='Centroids')
plt.xlabel("Sepal Length")
plt.ylabel("Sepal Width")
plt.title("Iris Clusters")
plt.legend()
plt.show()
```


Iris Clusters
Cluster 1
Cluster 2
Cluster 3
Centroids
Sepal Width
4.5
4.0
3.5
3.0
2.5
2.0